CSA4707 – DEEP LEARNING

CO1 – ASSESSMENT 2: ANALYTICAL PROBLEM SOLVING

Assessment Title: Analyse the Mathematics of a Diabetic Retinopathy Screening Model

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INTRODUCTION

RetinaCare is a tele-ophthalmology service designed to screen fundus images of diabetic patients in rural primary health centres. Before clinical deployment, the mathematical properties of the screening model must be evaluated to ensure reliable predictions, appropriate optimisation, accurate classification and safe clinical decisions.

This assessment analyses the mathematical foundations of RetinaCare using linear algebra, singular value decomposition, convex optimisation, information theory, Bayesian statistics, gradient descent, bias–variance decomposition and classification metrics. A final design challenge examines how low disease prevalence and unequal clinical error costs affect the screening policy.

The objective is to evaluate the strengths and limitations of the model and recommend an evidence-based screening strategy.

TASK 1 – BREAK DOWN THE AUDIT PROBLEM

1.1 Mathematical Decomposition

The RetinaCare audit consists of six supplied data items, each representing a mathematical component of a deep-learning system.

Data item	Mathematical concept	Model stage	Purpose
D1	Linear algebra and SVD	Feature representation	Measures the information retained during feature compression.
D2	Hessian, convexity and optimisation	Loss and optimisation	Examines curvature, the minimum and gradient-descent behaviour.
D3	Entropy, cross-entropy and KL divergence	Probability and loss evaluation	Measures the difference between the true and predicted distributions.
D4	Bayes' theorem	Screening decision	Determines the probability of disease given a positive or negative result.
D5	Bias–variance decomposition	Prediction evaluation	Separates systematic prediction error, variability and observation noise.
D6	Confusion matrix and classification metrics	Model evaluation	Quantifies correct classifications, false referrals and missed cases.

1.2 Dependencies Between Data Items

D1 and D2 analyse mathematical properties of the representation and optimisation process. D3 evaluates how closely the model's predicted probabilities match the true distribution. D4 and D6 are related because both investigate screening decisions, but D4 uses sensitivity, specificity and prevalence, whereas D6 uses observed confusion-matrix counts. D5 evaluates numerical prediction error independently of the classification counts.

The results of Tasks 2–9 contribute to the integrated audit in Task 10. Task 11 changes the prevalence and cost assumptions and therefore requires a new decision analysis.

1.3 Assumptions

1. The supplied matrices and probability distributions are treated as exact.
2. The screening sensitivity and specificity remain fixed unless an alternative operating point is explicitly considered.
3. The classification matrix represents 100 evaluated images.
4. The noise variance in D5 is assumed to be independent, zero-mean observation noise.
5. All information-theoretic logarithms use base 2.

6. Numerical answers are reported to four decimal places.

TASK 2 – EXAMINE THE COMPRESSIBILITY OF THE FEATURE MATRIX

2.1 Given Data

The feature matrix is

$A = [[3, 0], [4, 5]]$

The singular value decomposition is

$A = U\Sigma V^T$

The singular values are obtained from the eigenvalues of A^TA.

2.2 Calculate A^TA

$A^T = [[3, 4], [0, 5]]$

Therefore,

$A^TA = [[3, 4], [0, 5]]\ [[3, 0], [4, 5]]$

$A^TA = [[25, 20], [20, 25]]$

2.3 Find the Eigenvalues

The characteristic equation is

$\det(A^TA - \lambda I) = 0$

$|25 - \lambda, 20; 20, 25 - \lambda| = 0$

$(25 - \lambda)^2 - 400 = 0$

$25 - \lambda = \pm 20$

Hence,

$\lambda_1 = 45$

$\lambda_2 = 5$

The singular values are the square roots of the eigenvalues.

$\sigma_1 = \sqrt{45} = 6.7082$

$\sigma_2 = \sqrt{5} = 2.2361$

Therefore,

$\Sigma = [[6.7082, 0], [0, 2.2361]]$

2.4 Find the Right Singular Vectors

For $\lambda_1 = 45$,

$(A^TA - 45I)v_1 = 0$

This gives $-20x + 20y = 0$, so $x = y$.

The normalised eigenvector is

$v_1 = (1/\sqrt{2})[1, 1]^T$

For $\lambda_2 = 5$,

$(A^TA - 5I)v_2 = 0$

This gives $20x + 20y = 0$, so $x = -y$.

The normalised eigenvector is

$v_2 = (1/\sqrt{2})[1, -1]^T$

Therefore,

$V = (1/\sqrt{2})[[1, 1], [1, -1]]$

2.5 Find the Left Singular Vectors

The left singular vectors are calculated using

$u_i = Av_i/\sigma_i$

For the first singular vector,

$Av_1 = (1/\sqrt{2})[3, 9]^T$

$u_1 = [3/\sqrt{90}, 9/\sqrt{90}]^T$

Therefore,

$u_1 \approx [0.3162, 0.9487]^T$

For the second singular vector,

$Av_2 = (1/\sqrt{2})[3, -1]^T$

$u_2 = [3/\sqrt{10}, -1/\sqrt{10}]^T$

Therefore,

$u_2 \approx [0.9487, -0.3162]^T$

Thus, an SVD of A is

$U \approx [[0.3162, 0.9487], [0.9487, -0.3162]]$

$\Sigma \approx [[6.7082, 0], [0, 2.2361]]$

$V^T = (1/\sqrt{2})[[1, 1], [1, -1]]$

2.6 Compare the Singular Values

The singular-value ratio is

$\sigma_1/\sigma_2 = 6.7082/2.2361 = 3.0000$

The dominant singular value is three times the smaller singular value. This indicates that the first direction carries more information, but the second direction is not negligible.

2.7 Calculate the Rank-1 Approximation

The rank-1 approximation is

$A_1 = \sigma_1 u_1 v_1^T$

Using the derived vectors,

$A_1 \approx [[1.5000, 1.5000], [4.5000, 4.5000]]$

The reconstruction error is

$A - A_1 \approx [[1.5000, -1.5000], [-0.5000, 0.5000]]$

The squared Frobenius norm is

$||A - A_1||^2F = 1.5^2 + (-1.5)^2 + (-0.5)^2 + 0.5^2$

$= 2.25 + 2.25 + 0.25 + 0.25$

$= 5.0000$

Therefore,

$||A - A_1||F = \sqrt{5} = 2.2361$

2.8 Calculate the Energy Retained

The total energy is

$||A||^2F = 3^2 + 0^2 + 4^2 + 5^2$

$= 9 + 16 + 25 = 50$

The energy retained by the first singular component is

Energy retained $= \sigma_1^2/(\sigma_1^2 + \sigma_2^2) \times 100$

$= 45/(45 + 5) \times 100$

$= 90.0000\%$

The discarded energy is 10.0000%.

2.9 Interpretation

The dominant singular direction retains 90% of the total matrix energy, and the rank-1 reconstruction error is 2.2361. Therefore, rank-1 compression provides a substantial reduction in representation size. However, it removes 10% of the matrix energy. In medical screening, compression should only be accepted after verifying that the discarded information does not contain clinically important disease features.

TASK 3 – ANALYSE THE CURVATURE OF THE LOSS SURFACE

3.1 Given Data

$J(w) = \frac{1}{2}w^THw - b^Tw$

$H = [[4, 1], [1, 4]]$

$b = [5, 5]^T$

$w_0 = [0, 0]^T$

3.2 Calculate the Eigenvalues

The characteristic equation is

$\det(H - \lambda I) = 0$

$|4 - \lambda, 1; 1, 4 - \lambda| = 0$

$(4 - \lambda)^2 - 1 = 0$

$4 - \lambda = \pm 1$

Therefore,

$\lambda_1 = 5$

$\lambda_2 = 3$

3.3 Calculate the Eigenvectors

For $\lambda_1 = 5$, the eigenvector is proportional to $[1, 1]^T$.

The normalised eigenvector is

$v_1 = (1/\sqrt{2})[1, 1]^T$

For $\lambda_2 = 3$, the eigenvector is proportional to $[1, -1]^T$.

The normalised eigenvector is

$v_2 = (1/\sqrt{2})[1, -1]^T$

The direction [1, 1] has the higher curvature of 5, while [1, −1] has the lower curvature of 3.

3.4 Positive Definiteness and Convexity

The eigenvalues of H are 5 and 3. Both are positive.

Therefore, H is positive definite.

For a quadratic loss, a positive-definite Hessian implies that the loss function is strictly convex. Consequently, the loss has a unique global minimum.

3.5 Condition Number

For a symmetric positive-definite matrix,

$\kappa(H) = \lambda max/\lambda min$

$= 5/3$

= 1.6667

The relatively small condition number indicates that the curvature is not extremely uneven. Gradient descent should therefore avoid severe oscillations when an appropriate learning rate is selected.

3.6 Calculate the Minimiser

At the minimum,

$\nabla J(w) = Hw - b = 0$

Therefore,

$Hw^* = b$

$H^{-1} = (1/15)[[4, -1], [-1, 4]]$

Hence,

$w^* = H^{-1}b$

$= (1/15)[[4, -1], [-1, 4]][5, 5]^T$

$= (1/15)[15, 15]^T$

Therefore,

$w^* = [1, 1]^T$

Verification:

$\nabla J(w^*) = [[4, 1], [1, 4]][1, 1]^T - [5, 5]^T$

$= [5, 5]^T - [5, 5]^T$

$= [0, 0]^T$

The gradient is zero, confirming the minimiser.

3.7 Minimum Loss

$J(w^*) = ½[1, 1][5, 5]^T - [5, 5][1, 1]^T$

$= ½(10) - 10$

$= -5.0000$

3.8 Interpretation

The positive eigenvalues confirm a strictly convex loss surface with a unique minimum at $[1, 1]^T$. The condition number of 1.6667 indicates moderate curvature variation. These properties support the use of gradient-based optimisation, provided that the learning rate is chosen within a stable range.

TASK 4 – DIFFERENTIATE THE MODELS USING INFORMATION THEORY

4.1 Given Data

The true distribution is

$p = [0.5, 0.25, 0.125, 0.125]$

The untrained model predicts

$q_1 = [0.25, 0.25, 0.25, 0.25]$

The partly trained model predicts

$q_2 = [0.4, 0.3, 0.2, 0.1]$

4.2 Calculate the Entropy of p

Entropy is defined as

$H(p) = -\Sigma\, p_i \log_2 p_i$

Substituting,

$H(p) = -[0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.125 \log_2 0.125 + 0.125 \log_2 0.125]$

$H(p) = -[0.5(-1) + 0.25(-2) + 0.125(-3) + 0.125(-3)]$

$= 0.5 + 0.5 + 0.375 + 0.375$

$H(p) = 1.7500$ bits

4.3 Calculate $H(q_1)$

Since q_1 is uniform,

$H(q_1) = -4(0.25 \log_2 0.25)$

$= -4(0.25 \times -2)$

$H(q_1) = 2.0000$ bits

4.4 Calculate Cross-Entropy $H(p, q_1)$

Cross-entropy is

$H(p, q_1) = -\Sigma\, p_i \log_2 q_{1i}$

Because every q_{1i} equals 0.25,

$H(p, q_1) = -\log_2 0.25\, \Sigma p_i$

$= -(-2)(1)$

$H(p, q_1) = 2.0000$ bits

4.5 Verify the KL Divergence Decomposition

KL divergence is

$KL(p \,||\, q_1) = \Sigma p_i \log_2(p_i/q_{1i})$

Substituting,

$KL(p \,||\, q_1) = 0.5 \log_2 2 + 0.25 \log_2 1 + 0.125 \log_2 0.5 + 0.125 \log_2 0.5$

$= 0.5 + 0 - 0.125 - 0.125$

$= 0.2500$ bits

The cross-entropy identity is

$H(p, q_1) = H(p) + KL(p \,||\, q_1)$

$= 1.7500 + 0.2500$

$= 2.0000$ bits

This matches the directly calculated cross-entropy.

4.6 Calculate $KL(p \,||\, q_2)$

$KL(p \,||\, q_2) = \Sigma p_i \log_2(p_i/q_{2i})$

$= 0.5 \log_2(0.5/0.4) + 0.25 \log_2(0.25/0.3) + 0.125 \log_2(0.125/0.2) + 0.125 \log_2(0.125/0.1)$

The individual contributions are approximately:

Class	Calculation	Contribution (bits)
No DR	$0.5 \log_2(1.25)$	0.1610
Mild	$0.25 \log_2(0.8333)$	−0.0660
Moderate	$0.125 \log_2(0.625)$	−0.0845
Severe	$0.125 \log_2(1.25)$	0.0402

Adding the contributions using unrounded values,

$KL(p \parallel q_2) \approx 0.0500$ bits

More precisely, $KL(p \parallel q_2) \approx 0.0499$ bits.

4.7 Compare the Models

$KL(p \parallel q_1) = 0.2500$ bits

$KL(p \parallel q_2) \approx 0.0499$ bits

The partly trained model has the smaller divergence and therefore represents the true distribution more closely.

The No DR class contributes positively because q_2 underestimates its true probability. The Mild and Moderate classes contribute negatively individually, but KL divergence remains non-negative when all contributions are summed. The Severe class contributes positively because its predicted probability is lower than its true probability.

A divergence of zero means $p_i = q_i$ for every class with positive true probability. In that case, the model distribution matches the true distribution exactly.

4.8 Interpretation

The reduction in KL divergence from 0.2500 to approximately 0.0499 bits indicates improvement in probabilistic predictions. However, distributional similarity alone does not establish clinical safety. Sensitivity, missed-case rates and calibration must also be examined.

TASK 5 – ANALYSE THE SCREENING DECISION USING BAYES' RULE

5.1 Given Data

Prevalence = 0.10

Sensitivity = 0.90

Specificity = 0.95

False-positive rate = 1 − 0.95 = 0.05

False-negative rate = 1 − 0.90 = 0.10

5.2 Probability Tree

For 1,000 screened patients:

- Referable patients: 100
- Non-referable patients: 900
- True positives: 100 × 0.90 = 90
- False negatives: 100 × 0.10 = 10
- True negatives: 900 × 0.95 = 855
- False positives: 900 × 0.05 = 45

5.3 Positive Predictive Value

Bayes' theorem gives

$P(R \mid +) = P(+ \mid R)P(R)/P(+)$

$P(R \mid +) = (0.90 \times 0.10)/[(0.90 \times 0.10) + (0.05 \times 0.90)]$

$= 0.09/(0.09 + 0.045)$

$= 0.09/0.135$

$P(R \mid +) = 0.6667 = 66.6667\%$

5.4 Probability of Disease After a Negative Result

$P(R \mid -) = P(- \mid R)P(R)/P(-)$

$= (0.10 \times 0.10)/[(0.10 \times 0.10) + (0.95 \times 0.90)]$

$= 0.01/(0.01 + 0.855)$

$= 0.0116$

Therefore,

$P(R \mid -) = 1.1561\%$

The negative predictive value is

$NPV = 855/(855 + 10)$

$= 98.8439\%$

5.5 Interpretation

Sensitivity measures the proportion of referable patients correctly detected, while positive predictive value measures the probability that a patient with a positive result actually has referable disease. These are different quantities.

A positive result does not establish a definitive diagnosis. It should trigger appropriate clinical assessment and referral rather than automatic treatment. A negative result also does not completely eliminate disease risk.

TASK 6 – TRACE THE CONVERGENCE OF GRADIENT DESCENT

6.1 Gradient Formula

The loss function is

$J(w) = \frac{1}{2}w^T Hw - b^T w$

Differentiating,

$\nabla J(w) = Hw - b$

For the supplied matrix,

$\nabla J(w) = [4w_1 + w_2 - 5, w_1 + 4w_2 - 5]^T$

The gradient-descent update is

$w_{t+1} = w_t - \eta \nabla J(w_t)$

Given $\eta = 0.1$ and $w_0 = [0, 0]^T$.

6.2 Iteration 1

At $w_0 = [0, 0]^T$,

$\nabla J(w_0) = [-5, -5]^T$

$w_1 = [0, 0]^T - 0.1[-5, -5]^T$

$w_1 = [0.5000, 0.5000]^T$

Loss:

$J(w_1) = \frac{1}{2}(0.5 \times 5) - 5(0.5 + 0.5)$

$= 1.2500 - 5.0000$

$J(w_1) = -3.7500$

6.3 Iteration 2

At $w_1 = [0.5, 0.5]^T$,

$\nabla J(w_1) = [-2.5, -2.5]^T$

$w_2 = [0.5, 0.5]^T - 0.1[-2.5, -2.5]^T$

$w_2 = [0.7500, 0.7500]^T$

Loss:

$J(w_2) = \frac{1}{2}(0.75 \times 7.5) - 5(1.5)$

$= 2.8125 - 7.5000$

$J(w_2) = -4.6875$

6.4 Iteration 3

At $w_2 = [0.75, 0.75]^T$,

$\nabla J(w_2) = [-1.25, -1.25]^T$

$w_3 = [0.75, 0.75]^T - 0.1[-1.25, -1.25]^T$

$w_3 = [0.8750, 0.8750]^T$

Loss:

$J(w_3) = \frac{1}{2}(0.875 \times 8.75) - 5(1.75)$

$= 3.828125 - 8.7500$

$J(w_3) = -4.9219$

6.5 Distance to the Minimiser

The minimiser is $w^* = [1, 1]^T$.

Using Euclidean distance,

$\|w_t - w^*\|_2 = \sqrt{[(w_1 - 1)^2 + (w_2 - 1)^2]}$

Iteration	Weights	Distance to w*	Loss
0	(0, 0)	1.4142	0.0000
1	(0.5, 0.5)	0.7071	−3.7500
2	(0.75, 0.75)	0.3536	−4.6875
3	(0.875, 0.875)	0.1768	−4.9219

6.6 Interpretation

The distance to the minimiser is halved at each iteration. The loss approaches −5.0000 without oscillation.

The eigenvalues are 5 and 3. For $\eta = 0.1$, the error factors along the eigenvector directions are $1 - 0.1(5) = 0.5$ and $1 - 0.1(3) = 0.7$. The initial error lies entirely along the [1, 1] direction, which corresponds to eigenvalue 5. Therefore, its distance is halved at each step.

TASK 7 – DIFFERENTIATE MOMENTUM FROM PLAIN GRADIENT DESCENT

7.1 Momentum Update

Using the conventional velocity definition,

$v_{t+1} = \beta v_t + \nabla J(w_t)$

$w_{t+1} = w_t - \eta v_{t+1}$

Given $\beta = 0.9$, $\eta = 0.1$, $w_0 = [0, 0]^T$ and $v_0 = [0, 0]^T$.

7.2 First Momentum Iteration

$\nabla J(w_0) = [-5, -5]^T$

$v_1 = 0.9[0, 0]^T + [-5, -5]^T$

$v_1 = [-5, -5]^T$

$w_1 = [0, 0]^T - 0.1[-5, -5]^T$

$w_1 = [0.5000, 0.5000]^T$

7.3 Second Momentum Iteration

$\nabla J(w_1) = [-2.5, -2.5]^T$

$v_2 = 0.9[-5, -5]^T + [-2.5, -2.5]^T$

$v_2 = [-7, -7]^T$

$w_2 = [0.5, 0.5]^T - 0.1[-7, -7]^T$

$w_2 = [1.2000, 1.2000]^T$

The second update passes the minimiser at $[1, 1]^T$. Thus, momentum overshoots the minimum because the accumulated velocity continues moving in the previous direction.

For comparison, plain gradient descent gives $w_2 = [0.7500, 0.7500]^T$ after two iterations.

7.4 Learning-Rate Bound

For plain gradient descent on a positive-definite quadratic, convergence requires

$0 < \eta < 2/\lambda max$

Here $\lambda max = 5$.

Therefore,

$0 < \eta < 2/5$

$0 < \eta < 0.4000$

This bound applies to plain gradient descent. Momentum has its own stability conditions.

7.5 Why η = 0.5 Fails

Using plain gradient descent,

$w_1 = w_0 - 0.5\nabla J(w_0)$

$= [0, 0]^T - 0.5[-5, -5]^T$

$= [2.5, 2.5]^T$

The next gradient is

$\nabla J(w_1) = [7.5, 7.5]^T$

$w_2 = [2.5, 2.5]^T - 0.5[7.5, 7.5]^T$

$= [-1.25, -1.25]^T$

The iterates cross the minimum and move farther away. Along the eigenvalue-5 direction, the error factor is $1 - 0.5(5) = -1.5$, whose magnitude exceeds 1. Therefore, the error grows instead of shrinking.

7.6 Interpretation

Momentum can accelerate convergence but may overshoot when velocity accumulates. A suitable learning rate is essential. The $\eta = 0.5$ example violates the plain gradient-descent stability bound and produces divergence.

TASK 8 – DECOMPOSE THE PREDICTION ERROR

8.1 Given Data

True score $y = 10$

Noise variance $\sigma^2 = 1$

Model P predictions = [8, 9, 8, 9]

Model Q predictions = [12, 7, 11, 8]

The supplied predictions are treated as repeated predictions for the same target. Bias and variance are calculated using the population-variance convention.

8.2 Model P

Mean prediction:

$\bar{y}P = (8 + 9 + 8 + 9)/4$

$\bar{y}P = 8.5000$

Bias:

$BiasP = 8.5 - 10 = -1.5000$

Squared bias:

$BiasP^2 = 2.2500$

Variance:

$Var(P) = [(8 - 8.5)^2 + (9 - 8.5)^2 + (8 - 8.5)^2 + (9 - 8.5)^2]/4$

$= (0.25 + 0.25 + 0.25 + 0.25)/4$

$Var(P) = 0.2500$

Expected squared error:

$EPE(P) = Bias^2 + Variance + \sigma^2$

$= 2.25 + 0.25 + 1$

$EPE(P) = 3.5000$

8.3 Model Q

Mean prediction:

$\bar{y}Q = (12 + 7 + 11 + 8)/4$

$\bar{y}Q = 9.5000$

Bias:

$BiasQ = 9.5 - 10 = -0.5000$

Squared bias:

$BiasQ^2 = 0.2500$

Variance:

$\text{Var}(Q) = [(12 - 9.5)^2 + (7 - 9.5)^2 + (11 - 9.5)^2 + (8 - 9.5)^2]/4$

$= (6.25 + 6.25 + 2.25 + 2.25)/4$

$= 17/4$

$\text{Var}(Q) = 4.2500$

Expected squared error:

$\text{EPE}(Q) = 0.25 + 4.25 + 1$

$\text{EPE}(Q) = 5.5000$

8.4 Comparison

Component	Model P	Model Q
Mean prediction	8.5000	9.5000
Bias	−1.5000	−0.5000
Bias²	2.2500	0.2500
Variance	0.2500	4.2500
Noise variance	1.0000	1.0000
Expected squared error	3.5000	5.5000

Model P has the larger squared bias, whereas Model Q has the larger variance. The high variability of Model Q dominates its error decomposition.

Model P is preferable under the given expected squared-error criterion because 3.5000 is lower than 5.5000. This comparison concerns numerical prediction error and does not by itself establish which classifier is safer for clinical screening.

TASK 9 – EXAMINE THE ERROR PROFILE OF THE CLASSIFIER

9.1 Given Confusion Matrix

Actual class	Predicted referable	Predicted non-referable	Total
Actual referable	TP = 40	FN = 10	50

Actual non-referable	FP = 5	TN = 45	50
Total	45	55	100

9.2 Accuracy

Accuracy = (TP + TN)/(TP + TN + FP + FN)

= (40 + 45)/100

= 0.8500

Accuracy = 85.0000%

9.3 Error Rate

Error rate = (FP + FN)/Total

= (5 + 10)/100

= 0.1500

Error rate = 15.0000%

9.4 Precision

Precision = TP/(TP + FP)

= 40/(40 + 5)

= 0.8889

Precision = 88.8889%

9.5 Recall or Sensitivity

Recall = TP/(TP + FN)

= 40/(40 + 10)

= 0.8000

Recall = 80.0000%

9.6 Specificity

Specificity = TN/(TN + FP)

= 45/(45 + 5)

= 0.9000

Specificity = 90.0000%

9.7 F1-Score

F1 = 2 × Precision × Recall/(Precision + Recall)

Alternatively,

F1 = 2TP/(2TP + FP + FN)

= 80/(80 + 5 + 10)

= 80/95

F1 = 0.8421

F1-score = 84.2105%

9.8 Interpretation

The classifier misses 10 referable cases and incorrectly refers 5 non-referable cases. False negatives are twice as frequent as false positives.

A missed referable case can delay ophthalmological assessment and treatment, potentially increasing the risk of vision loss. A false referral can create unnecessary travel, expense and workload for rural patients and healthcare staff.

Accuracy alone is incomplete because it does not distinguish the clinical consequences of the two error types. Recall and the false-negative count must also be considered.

TASK 10 – RELATE THE FINDINGS TO THE AUDIT QUESTION

10.1 Strengths of the Mathematical Design

The SVD analysis retains 90.0000% of the matrix energy using rank-1 compression. This supports compression as a potential efficiency improvement, although clinical information loss must be validated.

The Hessian has positive eigenvalues 5 and 3, and the loss has a unique minimum at $[1, 1]^T$. Its condition number is 1.6667, supporting stable optimisation with an appropriate learning rate.

The partly trained model has KL divergence of approximately 0.0499 bits, compared with 0.2500 bits for the uniform model. This indicates that its probability distribution is closer to the true distribution.

Gradient descent with $\eta = 0.1$ reduces the distance to the minimiser from 1.4142 to 0.1768 after three iterations.

10.2 Concerns About Screening Errors

The classifier has 85.0000% accuracy, 80.0000% recall and 90.0000% specificity. It misses 10 of the 50 referable cases in the supplied confusion matrix.

The Bayesian analysis also demonstrates that predictive values depend on prevalence. At 10% prevalence, the positive predictive value is 66.6667%, which means that a substantial fraction of positive results may not represent referable disease.

10.3 Audit Conclusion

The supplied mathematics supports the basic use of SVD, a convex quadratic loss and gradient-based optimisation. However, these properties do not guarantee clinical safety.

The 80.0000% recall and the 10 observed false negatives raise concerns about missed disease. The audit committee should require validation on representative fundus images, subgroup evaluation, threshold analysis, calibration checks and clinical review before approving deployment.

TASK 11 – DAY-4 DESIGN CHALLENGE: ANALYSE THE IMPACT OF LOW PREVALENCE

A. Impact Analysis

The district prevalence of referable retinopathy decreases from 10% to 2%. Sensitivity remains 90%, and specificity remains 95%.

Prevalence affects the prior probability of disease and the posterior probabilities calculated using Bayes' theorem. Sensitivity and specificity are conditional performance measures and are assumed unchanged in this scenario.

The SVD results, Hessian eigenvalues, loss minimiser, gradient-descent calculations and D5 numerical prediction decomposition do not change because their supplied inputs remain the same.

The predictive values change because the proportions of referable and non-referable patients in the screened population change.

B. Recalculate Probabilities and Expected Counts

B.1 Expected Counts at 2% Prevalence

Consider 1,000 patients.

Referable patients = 1,000 × 0.02 = 20

Non-referable patients = 1,000 − 20 = 980

True positives:

TP = 20 × 0.90 = 18

False negatives:

FN = 20 × 0.10 = 2

True negatives:

TN = 980 × 0.95 = 931

False positives:

FP = 980

Measure	**10% prevalence**	**2% prevalence**
Referable patients	100	20
Non-referable patients	900	980
True positives	90	18
False negatives	10	2
True negatives	855	931
False positives	45	49
Total positive results	135	67
Total negative results	865	933

The number of missed cases decreases from 10 to 2 per 1,000 patients because fewer patients have referable disease. The number of false referrals increases from 45 to 49 because there are more non-referable patients exposed to the 5% × 0.05 = 49

false-positive rate.

B.2 Positive Predictive Value at 2% Prevalence

PPV = TP/(TP + FP)

= 18/(18 + 49)

= 18/67

PPV = 0.2687

Positive predictive value = 26.8657%

B.3 Probability of Disease After a Negative Result

P(R | −) = FN/(FN + TN)

= 2/(2 + 931)

= 2/933

P(R | −) = 0.0021

Therefore, the probability of referable disease after a negative result is approximately 0.2144%.

The negative predictive value is

NPV = 931/933

= 0.9979

NPV = 99.7856%

B.4 Interpretation

At 2% prevalence, only 18 of the 67 positive results are expected to be true positives. Therefore, a positive result has a lower predictive value than at 10% prevalence.

The negative predictive value is high, but two referable cases per 1,000 screened patients are still expected to be missed under the stated assumptions.

C. Analyse the Screening Policy Options

C.1 Minimum Specificity for PPV of at Least 50%

The requirement is

$PPV \geq 0.5$

Using Bayes' theorem,

$PPV = Se \times Prev/[Se \times Prev + (1 - Sp)(1 - Prev)]$

Substitute sensitivity = 0.90 and prevalence = 0.02.

$0.5 \leq (0.90 \times 0.02)/[(0.90 \times 0.02) + (1 - Sp)(0.98)]$

$0.5 \leq 0.018/[0.018 + 0.98(1 - Sp)]$

For PPV to be at least 0.5, the false-positive contribution must not exceed the true-positive contribution.

$0.98(1 - Sp) \leq 0.018$

$1 - Sp \leq 0.018/0.98$

$1 - Sp \leq 0.0183673$

$Sp \geq 0.9816327$

Therefore, the minimum specificity is

$Sp \geq 98.1633\%$

This requirement assumes sensitivity remains at 90%. In practice, improving specificity by changing a model's threshold may reduce sensitivity.

C.2 Effect of Changing the Decision Threshold

Lowering the referral threshold generally increases sensitivity and reduces missed cases, but it can decrease specificity and increase false referrals.

Raising the referral threshold generally increases specificity and reduces false referrals, but it can decrease sensitivity and increase missed cases.

Because missing a referable case is ten times as costly as an unnecessary referral, a threshold should not be selected based on PPV alone. The decision must minimise the expected clinical cost while maintaining acceptable detection performance.

C.3 Expected Cost per 1,000 Patients

Let the cost of one false referral be 1 unit and the cost of one missed referable case be 10 units.

Expected cost = 10 × FN + FP

Under the original 10% prevalence:

Expected cost = 10(10) + 45

= 145 units per 1,000 patients.

Under the 2% prevalence with the original screening policy:

Expected cost = 10(2) + 49

= 69 units per 1,000 patients.

Although the total expected cost decreases, the number of false referrals exceeds the number of true referrals in the low-prevalence scenario.

For a hypothetical alternative operating point with sensitivity 90% and specificity 98.1633%, the expected counts are:

TP = 20 × 0.90 = 18

FN = 20 × 0.10 = 2

FP = 980 × (1 − 0.9816327) ≈ 18

Expected cost = 10(2) + 18

= approximately 38 units per 1,000 patients.

Policy scenario	**Sensitivity**	**Specificity**	**Expected FN**	**Expected FP**	**Expected cost**
Original policy at 10% prevalence	90%	95%	10	45	145

Original policy at 2% prevalence	90%	95%	2	49	69
Hypothetical improved-specificity policy at 2%	90%	98.1633%	2	Approximately 18	Approximately 38

The last row is a mathematical scenario, not a demonstrated model result. It assumes that specificity can be improved without changing sensitivity, which must be validated using actual screening data.

C.4 Limitations of the Cost Comparison

The numerical costs treat one unnecessary referral as one cost unit and one missed case as ten units. They do not include model development costs, travel expenses, follow-up capacity, disease severity or differences in treatment outcomes.

The cost comparison is therefore useful for policy analysis but is not a complete clinical or economic evaluation.

CONCLUSION

The mathematical audit demonstrates that the behaviour of a screening model depends on more than optimisation accuracy alone. SVD shows that rank-1 compression retains 90% of the supplied matrix energy, while the positive-definite Hessian guarantees a unique minimum for the quadratic loss. Information-theoretic analysis shows that the partly trained distribution is closer to the true distribution than the uniform model.

Gradient descent converges with a suitable learning rate, whereas an excessively large learning rate causes divergence. Bias–variance analysis identifies lower expected squared error for Model P, and the supplied confusion matrix reveals an 80% recall rate with 10 false negatives.

The low-prevalence design challenge demonstrates that predictive values can change substantially even when sensitivity and specificity remain fixed. Under the specified cost ratio, reducing unnecessary referrals is important, but missed referable cases must remain a central safety concern.

The final recommendation is to modify the screening policy through validated threshold selection, clinical confirmation and continued monitoring. The system should be approved for clinical deployment only after its performance and safety have been demonstrated on representative patient data.